1 The tensor product and the Kronecker product as one of its possible representations

1.1 The tensor product of vector spaces

Let $\underline{v} = \{v_i \mid i \in I\}$ and $\underline{w} = \{w_j \mid j \in J\}$ be bases for V and W. Then $V \otimes W$ is the product space, where there exists a basis which can be identified uniquely with the pairs of the cartesian product

$$\underline{v} \times \underline{w} = \big\{(v_i, w_j) \mid i \in I,\ j \in J\big\}, \tag{1}$$

so that

$$\dim(V \otimes W) = \dim V \cdot \dim W. \tag{2}$$

Since $V \otimes W$ is a vector space, an arbitrary element of it can be written as

$$\sum_{(i,j)\in I\times J} c_{ij}\,(v_i \otimes w_j). \tag{3}$$

The bilinear map $\otimes$ is defined by

$$\otimes\colon V \times W \longrightarrow V \otimes W, \tag{4}$$

$$(v, w) \longmapsto v \otimes w = \sum_{(i,j)\in I\times J} a_i b_j\,(v_i \otimes w_j), \tag{5}$$

for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$. As a check of bilinearity, writing $v' = \sum_i a'_i v_i$ and $v'' = \sum_i a''_i v_i$,

$$\begin{aligned}(v' + v'') \otimes w &= \Big(\textstyle\sum_i a'_i v_i + \sum_i a''_i v_i\Big) \otimes w = \Big(\sum_i (a'_i + a''_i)\, v_i\Big) \otimes w \\ &= \sum_{i,j} (a'_i + a''_i)\, b_j\,(v_i \otimes w_j) \\ &= \sum_{i,j} a'_i b_j\,(v_i \otimes w_j) + \sum_{i,j} a''_i b_j\,(v_i \otimes w_j) \\ &= v' \otimes w + v'' \otimes w,\end{aligned} \tag{6}$$

and analogously in the second argument.

1.2 The tensor product of operators

Introduce operators $A\colon V \to V$ and $B\colon W \to W$ with V, W vector spaces. How is $A \otimes B\colon V \otimes W \to V \otimes W$ given? Define a map $\beta\colon V \times W \to V \otimes W$, $\beta(v, w) := Av \otimes Bw$. It is bilinear, e.g.

$$\begin{aligned}\beta(v_1 + v_2, w) &= A(v_1 + v_2) \otimes Bw = (Av_1 + Av_2) \otimes Bw \\ &\overset{\otimes \text{ bilin.}}{=} Av_1 \otimes Bw + Av_2 \otimes Bw = \beta(v_1, w) + \beta(v_2, w),\end{aligned} \tag{7}$$

and analogously in the second argument. By the universal property there exists a unique linear $\tilde{\beta}$ with $\tilde{\beta}(v \otimes w) := \beta(v, w)$, so that

$$(A \otimes B)(v \otimes w) = Av \otimes Bw. \tag{8}$$

1.3 The Kronecker product as a matrix representation

Let $V = K^m$ and $W = K^n$ and identify (this is *not* a unique representation!) $K^m \times K^n \to K^{(m \cdot n)}$ with the Kronecker product, i.e. for

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \quad \& \quad w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \qquad \text{set} \qquad v \otimes w := \begin{pmatrix} a_1\, w \\ \vdots \\ a_m\, w \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ \vdots \\ a_1 b_n \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_n \end{pmatrix}. \tag{9}$$

Now choose $\{v_i\}$ to be the standard basis, so that $Av_i = A_{\cdot i}$ (the i-th column of A) and likewise for B. Then

$$\begin{aligned} (A \otimes B)(v_i \otimes v_j) = (Av_i) \otimes (Bv_j) &\overset{\text{std. basis}}{=} A_{\cdot i} \otimes B_{\cdot j} \\ = \begin{pmatrix} A_{1i}\, B_{\cdot j} \\ \vdots \\ A_{mi}\, B_{\cdot j} \end{pmatrix} &= \begin{pmatrix} \begin{pmatrix} A_{1i}B_{1j} \\ \vdots \\ A_{1i}B_{nj} \end{pmatrix} \\ \vdots \\ \begin{pmatrix} A_{mi}B_{1j} \\ \vdots \\ A_{mi}B_{nj} \end{pmatrix} \end{pmatrix}. \end{aligned} \tag{10}$$

Setting $v_i \otimes v_j \equiv w_k$ with $k = j + n\, i$ (for zero-based i), the matrix of $A \otimes B$ is the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} A_{11}B_{\cdot 1} & \cdots & A_{11}B_{\cdot n} & \cdots & A_{1m}B_{\cdot 1} & \cdots & A_{1m}B_{\cdot n} \\ \vdots & & & & & & \vdots \\ A_{m1}B_{\cdot 1} & \cdots & A_{m1}B_{\cdot n} & \cdots & A_{mm}B_{\cdot 1} & \cdots & A_{mm}B_{\cdot n} \end{pmatrix} \\ &= \begin{pmatrix} A_{11}B & \cdots & A_{1m}B \\ \vdots & \ddots & \vdots \\ A_{m1}B & \cdots & A_{mm}B \end{pmatrix}, \end{aligned} \tag{11}$$

which is the Kronecker product for matrices.

1.4 References

- *Tensorprodukt* and *Kronecker-Produkt*, German Wikipedia.